

Coding & Solution

Date	11 July 2026
Team ID	SWTID-2026-1298
Project Name	Smart Lender
Maximum Marks	5 Marks

Coding & Solution :

Solution Summary

Field	Details
Repository Link / URL	https://github.com/harshzxx/SmartLender-Loan-Prediction
Programming Language(s)	Python, HTML5, CSS3
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Key Features Implemented	• Loan eligibility prediction using Machine Learning. • Data preprocessing including handling missing values, encoding categorical features, and feature scaling. • Exploratory Data Analysis using Univariate, Bivariate, and Multivariate techniques. • Implementation and evaluation of Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost models. • Selection of the best-performing model based on evaluation metrics. • Flask-based web application for loan eligibility prediction. • Responsive user interface developed using HTML5 and CSS3. • .Model serialization using Pickle for deployment.
Pending / Incomplete Features	• User authentication and role-based access control can be added in future. • Prediction history and database integration can be enhanced. • Cloud deployment and real-time API integration are planned as future improvements.
Setup / Run Instructions	○ Clone the repository from GitHub. ○ Install the required Python libraries using <code>pip install -r requirements.txt</code>. ○ Execute the training notebook or script to generate the trained model (if required). ○ Run <code>python app.py</code> to start the Flask application. ○ Open the generated local URL in a web browser. ○ Enter applicant details and click Predict to view the loan eligibility result.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions/modules	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation where necessary	Yes
4	Error handling is implemented for critical operations	Partial
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The Smart Lender application has been developed using a modular approach, enabling each component to function independently while maintaining smooth integration. The machine learning workflow, from data preprocessing to prediction, has been implemented and validated successfully. The project is designed to support future enhancements such as user authentication, cloud deployment, and advanced reporting without requiring major architectural changes.